

ZOMATO RESTAURANT & FOOD DELIVERY ANALYSIS

1. Project Overview and Objective

The food delivery industry generates large volumes of data from restaurants, customers, and orders. However, raw operational data often contains data quality issues such as missing values, duplicate records, inconsistent formats, spelling variations, mixed data types, and invalid values. These issues can affect the accuracy of business reporting and decision-making.

The objective of this project is to clean, transform, and analyze a fictional Zomato-style restaurant and food delivery dataset for major Indian cities using Excel and Power BI. The project focuses on improving data quality, building relationships between restaurant, customer, order, and location data, and creating an interactive dashboard to generate meaningful business insights.

2. Data Sources

- Source Description and Timeline:** The dataset was AI-generated and is completely fictional, created specifically for learning, data analysis, and portfolio purposes **and** The Orders dataset covers transactions from **January 2023 to August 2026**.
- Domain:** E-Commerce

3. Problem Statement

- To analyze restaurant performance across cities and cuisines to identify top-performing restaurants and key rating trends.
- To study monthly and yearly order patterns to understand sales trends and overall revenue performance.
- To evaluate customer ordering behavior, payment preferences, and satisfaction to support better customer engagement strategies.
- To analyze delivery time, discounts, and order patterns to identify factors that impact customer satisfaction and business performance.

4. Attribute (Column /Features) Details:

4.1 Restaurants Table

Attribute Name	Data Type	Description
Restaurant_ID	String	Unique identifier for each restaurant
Restaurant_Name	String (Text)	Name of the restaurant

City	Categorical (Text)	City where the restaurant is located
Area	String (Text)	Area or locality of the restaurant
Address	String (Text)	Full address of the restaurant
Cuisine	Categorical (Text)	Type of cuisine offered by the restaurant
Restaurant_Type	Categorical (Text)	Type of restaurant, such as Café, Fine Dining, or Quick Bites
Average_Rating	Numeric (Decimal / Text)	Average rating of the restaurant
Total_Reviews	Numeric (Integer / Text)	Total number of customer reviews
Average_Cost_for_Two	Numeric / Currency / Text	Average cost for two people
Online_Order_Available	Categorical (Yes/No)	Indicates whether online ordering is available
Table_Booking_Available	Categorical (Yes/No)	Indicates whether table booking is available
Delivery_Available	Categorical (Yes/No)	Indicates whether food delivery is available
Opening_Year	Numeric (Integer)	Year in which the restaurant was opened
Restaurant_Status	Categorical (Text)	Current operational status of the restaurant

4.2 Orders Table

Attribute Name	Data Type	Description
Order_ID	String	Unique identifier for each order
Restaurant_ID	String	Identifier linking the order to a restaurant
Customer_ID	String	Identifier linking the order to a customer
Order_Date	Date / Text	Date on which the order was placed
Order_Time	Time / Text	Time at which the order was placed
Order_Amount	Numeric / Currency / Text	Original amount of the order before charges and discounts
Delivery_Fee	Numeric / Currency / Text	Delivery charge applied to the order
Discount_Amount	Numeric / Currency / Text	Discount provided on the order
Final_Order_Amount	Numeric / Currency / Text	Final amount paid by the customer
Payment_Method	Categorical (Text)	Method used to make the payment
Order_Status	Categorical (Text)	Current status of the order
Delivery_Time_Minutes	Numeric (Integer / Text)	Time taken to deliver the order in minutes
Customer_Rating	Numeric (Decimal / Text)	Rating provided by the customer

4.3 Customers Table

Attribute Name	Data Type	Description
Customer_ID	String	Unique identifier for each customer
Customer_Name	String (Text)	Name of the customer
City	Categorical (Text)	City where the customer is located
Age	Numeric (Integer)	Age of the customer in years
Gender	Categorical (Text)	Gender of the customer
Customer_Segment	Categorical (Text)	Customer category based on ordering behavior
Registration_Date	Date / Text	Date on which the customer registered
Total_Orders	Numeric (Integer / Text)	Total number of orders placed by the customer
Preferred_Cuisine	Categorical (Text)	Cuisine most preferred by the customer

4.4 Categories_Locations Table

Attribute Name	Data Type	Description
Location_ID	String	Unique identifier for each location record
City	Categorical (Text)	City where the restaurant is located
Area	String (Text)	Specific area or locality within the city
Zone	Categorical (Text)	Geographical zone, such as North, South, East, West, or Central
Restaurant_ID	String	Identifier linking the location record to a restaurant
Cuisine_Category	Categorical (Text)	Cuisine category associated with the restaurant
Price_Category	Categorical (Text)	Price classification, such as Budget, Mid-Range, or Premium

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.

The screenshot shows a Microsoft Excel spreadsheet titled 'Zomato_Analysis_Raw_Data - Excel'. A dialog box titled 'Microsoft Excel' is displayed in the center, stating: '25 duplicate values found and removed; 4200 unique values remain. Note that counts may include empty cells, spaces, etc.' The spreadsheet contains columns for Order_ID, Restaurant_ID, Customer_ID, Order_Date, Order_Time, Order_Amount, Delivery_Fee, Discount_Amount, Final_Order_Amount, Payment_Method, Order_Status, and Delivery_Time. The data is organized into rows, with some cells containing 'N/A' or '0'.

Handling null values using AVERAGE, AVERAGEIF and MODE

The screenshot shows a Microsoft Excel spreadsheet titled 'Zomato_Analysis_Cleaning - Excel'. A PivotTable is visible, showing data for Restaurants. The PivotTable has columns for Restaurant_ID, Restaurant_Name, City, Area, Address, Cuisine, and Restaurant_Type. The data is organized into rows, with some cells containing 'null' or '0'. A 'Query Settings' dialog box is open on the right, showing the 'Source' and 'Changed Type' options. The spreadsheet also includes a 'Restaurants' table with columns for Restaurant_ID, Restaurant_Name, City, Area, Address, Cuisine, and Restaurant_Type.

- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

Manage relationships

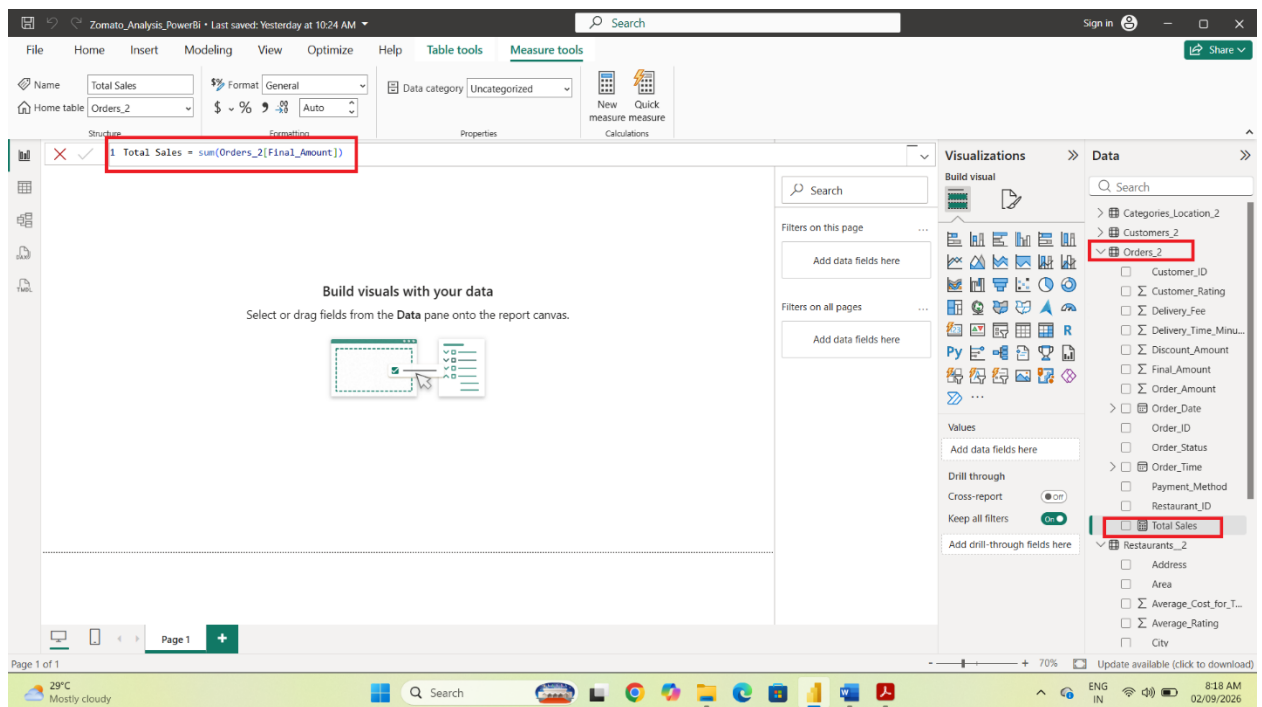
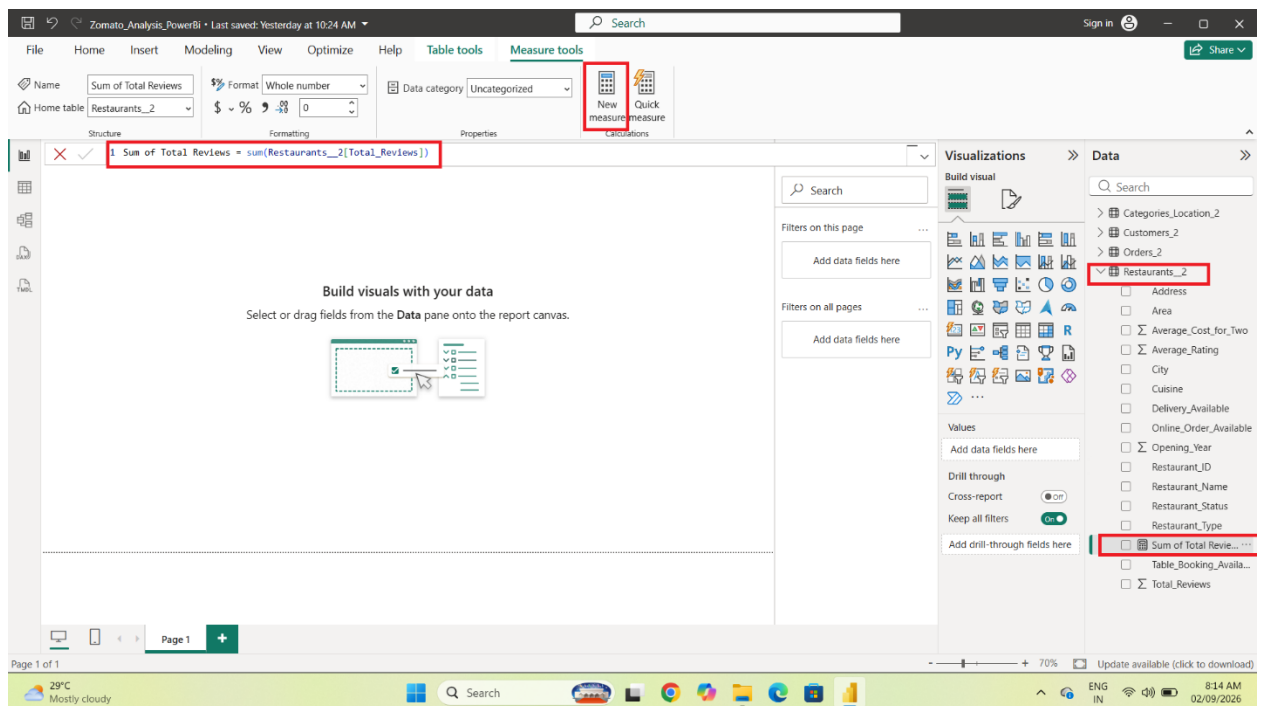
+ New relationship		⚡ Autodetect		Edit		Delete		Filter ▼	
☐	From: table (column) ↑	Relationship	To: table (column)	Status					
☐	Categories_Location_2 (Restau...	* ◀ 1	Restaurants_2 (Restaurant_ID)	Active ...					
☐	Orders_2 (Customer_ID)	* ◀ 1	Customers_2 (Customer_ID)	Active ...					
☐	Orders_2 (Restaurant_ID)	* ◀ 1	Restaurants_2 (Restaurant_ID)	Active ...					

Close

Calculated Columns & DAX Measures:

Implemented DAX formulas are listed below

DAX Measures:



Power BI Desktop interface showing the formula bar with the measure: $\text{Total Orders} = \text{DISTINCTCOUNT}(\text{Orders}_2[\text{Order_ID}])$. The Data pane on the right shows the 'Orders_2' table selected, with 'Total Orders' highlighted in the Values section. The Visualizations pane shows the 'Total' icon selected. The status bar at the bottom indicates 'Page 1 of 1' and '70%' zoom.

Power BI Desktop interface showing the formula bar with the measure: $\text{Average Order Value} = \text{DIVIDE}([\text{Total Sales}], [\text{Total Orders}], 0)$. The Data pane on the right shows the 'Orders_2' table selected, with 'Average Order Value' highlighted in the Values section. The Visualizations pane shows the 'Total' icon selected. The status bar at the bottom indicates 'Page 1 of 1' and '70%' zoom.

Microsoft Power BI Desktop interface showing the creation of a measure.

Measure Name: Total Discount

Formula: `Total Discount = sum(Orders_2[Discount_Amount])`

Data Source: Orders_2

Visualizations: The right-hand pane shows the available data fields. The field **Total Discount** is highlighted in the list.

Build visuals with your data: The central canvas displays a message: "Select or drag fields from the Data pane onto the report canvas."

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Microsoft Power BI Desktop interface showing the creation of a measure.

Measure Name: Average Delivery Time

Formula: `Average Delivery Time = AVERAGE(Orders_2[Delivery_Time_Minutes])`

Data Source: Orders_2

Visualizations: The right-hand pane shows the available data fields. The field **Average Delivery Time** is highlighted in the list.

Build visuals with your data: The central canvas displays a message: "Select or drag fields from the Data pane onto the report canvas."

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Power BI Desktop interface showing the creation of a measure named "Average Age".

Measure Name: Average Age

Formula: `AVERAGE(Customers_2[Age])`

Visualizations: The "Visualizations" pane on the right shows the "Average Age" measure selected under the "Values" section.

Data: The "Data" pane on the right shows the "Customers_2" table selected, with fields like "City", "Customer_ID", "Customer_Name", "Customer_Segment", "Gender", "Preferred_Cuisine", "Registration_Date", "Total_Orders", "Orders_2", and "Restaurants_2" listed.

Build visuals with your data: The central canvas displays the instruction "Select or drag fields from the Data pane onto the report canvas." with a visual guide showing a table icon and a chart icon.

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Power BI Desktop interface showing the creation of a measure named "Total Customers".

Measure Name: Total Customers

Formula: `counta(Customers_2[Customer_ID])`

Visualizations: The "Visualizations" pane on the right shows the "Total Customers" measure selected under the "Values" section.

Data: The "Data" pane on the right shows the "Customers_2" table selected, with fields like "City", "Customer_ID", "Customer_Name", "Customer_Segment", "Gender", "Preferred_Cuisine", "Registration_Date", "Total_Orders", "Orders_2", and "Restaurants_2" listed.

Build visuals with your data: The central canvas displays the instruction "Select or drag fields from the Data pane onto the report canvas." with a visual guide showing a table icon and a chart icon.

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Power BI Desktop interface showing the creation of a measure.

Measure Name: Sum of Total Orders
Formula: `sum(Customers_2[Total_Orders])`

Visualizations: The 'Visualizations' pane on the right shows the 'Sum of Total Orders' measure selected under the 'Values' section.

Data: The 'Data' pane on the right shows the 'Customers_2' table selected, with the 'Sum of Total Orders' measure highlighted.

Build visuals with your data: The central canvas displays the instruction: "Select or drag fields from the Data pane onto the report canvas."

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Power BI Desktop interface showing the creation of a measure.

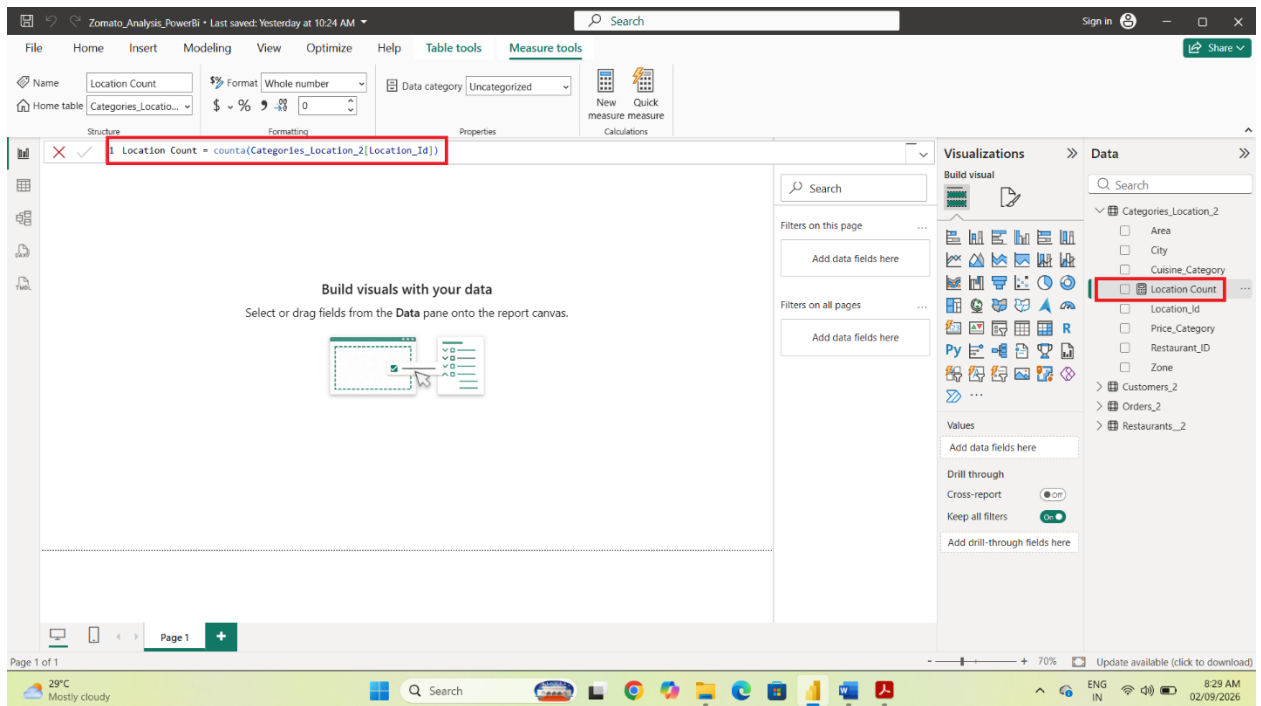
Measure Name: Location Count
Formula: `counta(Categories_Location_2[Location_Id])`

Visualizations: The 'Visualizations' pane on the right shows the 'Location Count' measure selected under the 'Values' section.

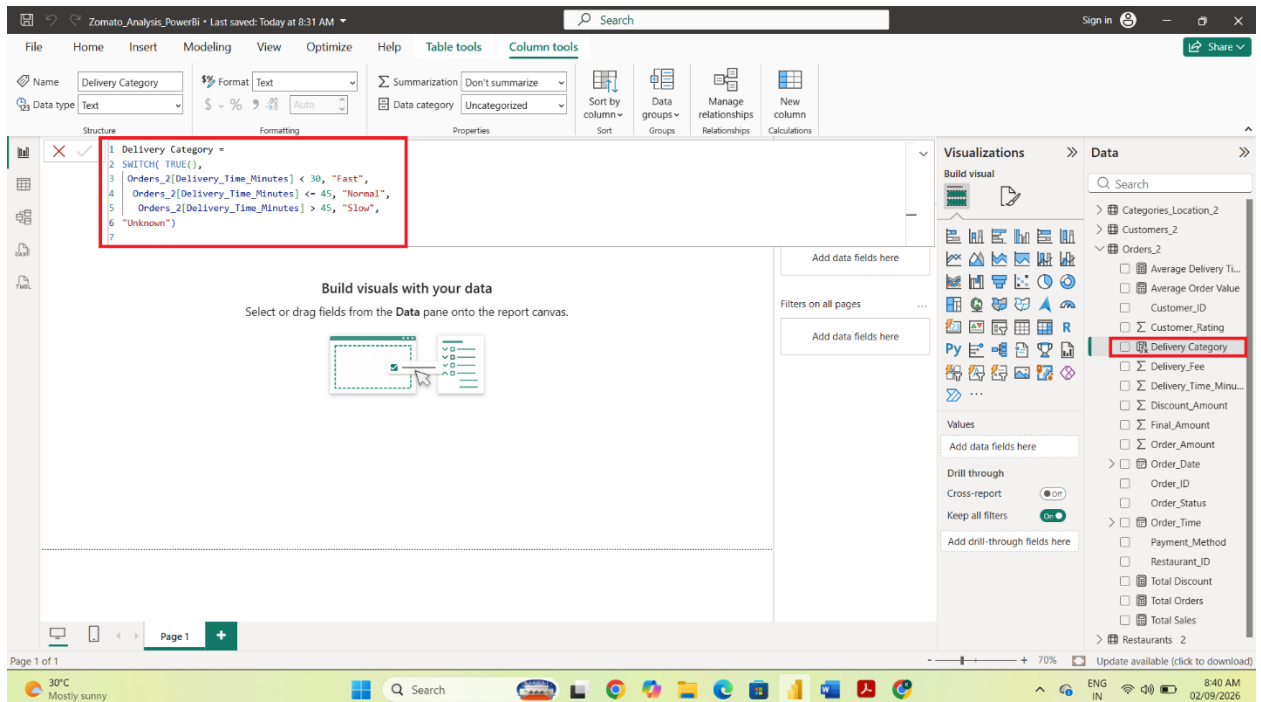
Data: The 'Data' pane on the right shows the 'Categories_Location_2' table selected, with the 'Location Count' measure highlighted.

Build visuals with your data: The central canvas displays the instruction: "Select or drag fields from the Data pane onto the report canvas."

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DAX Column:

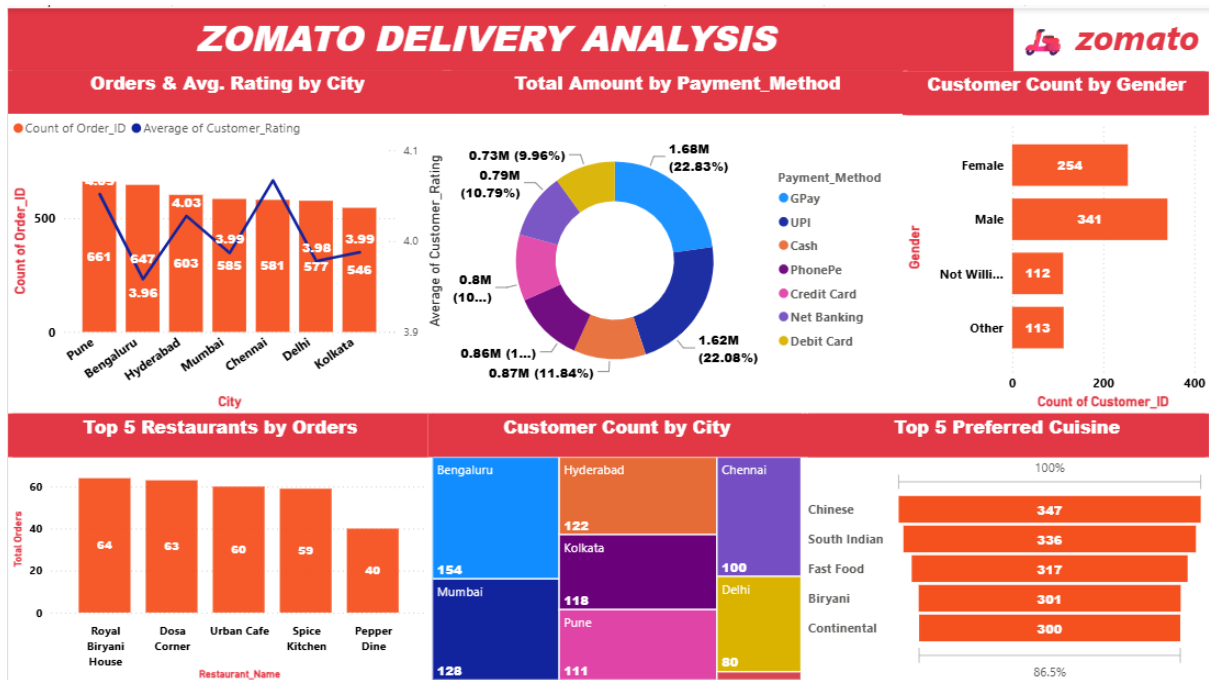
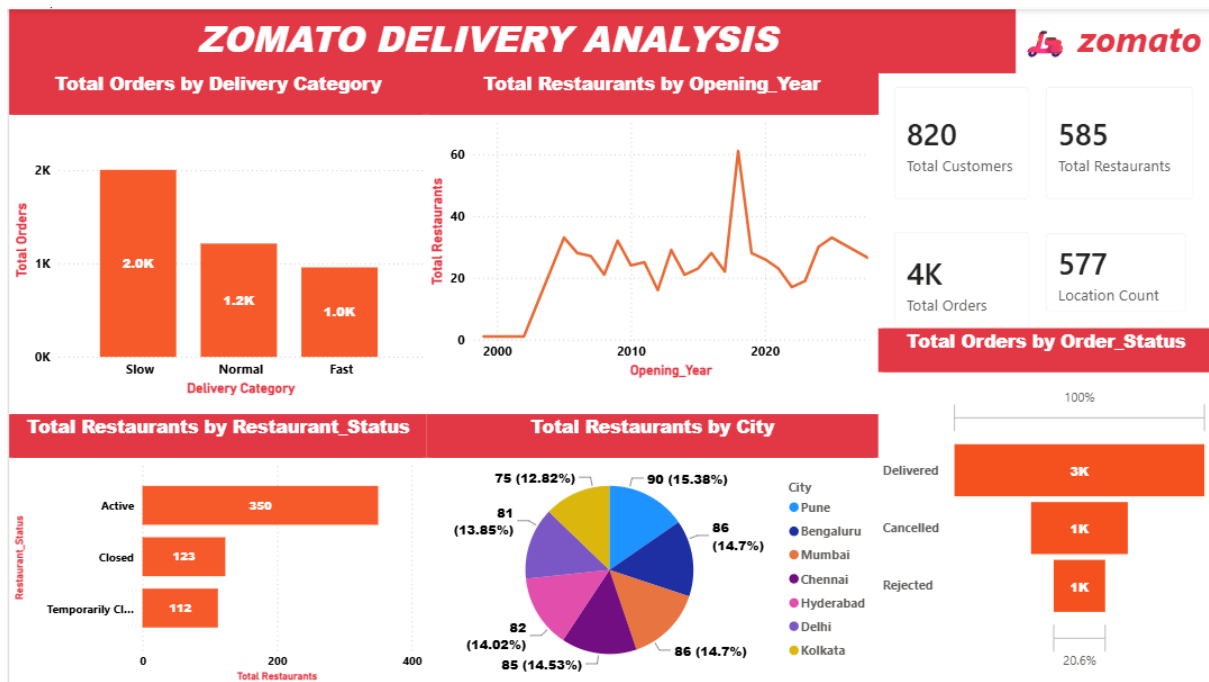


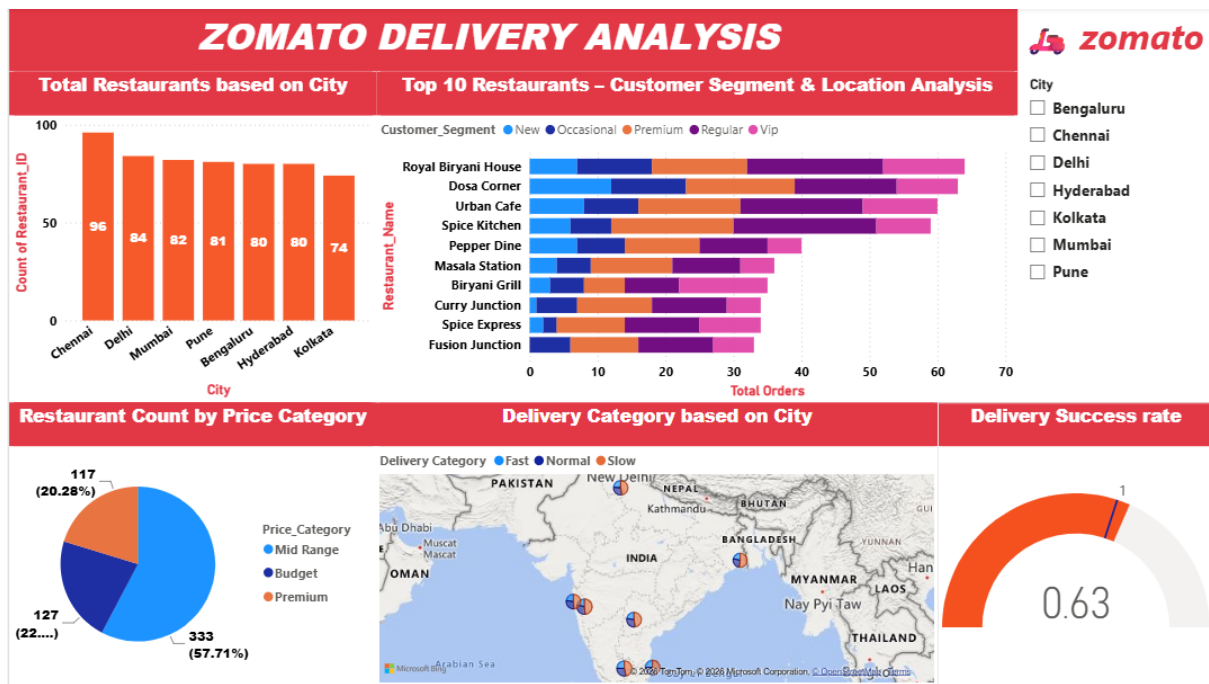
8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Multiple Visualizations based on problem statement: Bar charts, line charts, pie/donut charts, combo charts, funnel charts, treemaps, maps, gauge charts, and KPI cards to communicate business insights.

- **KPI Cards:** Displays key metrics such as Total Customers, Total Restaurants, Total Orders, and Location Count.
- **Restaurant Performance Analysis:** Analyzes restaurant count by city, restaurant status, opening year, and top restaurants by order volume.
- **Order & Delivery Analysis:** Provides insights into delivery categories, order status, city-wise orders, and delivery success rate.
- **Customer Analysis:** Analyzes customer count by city, gender, customer segment, and preferred cuisine.
- **Payment Analysis:** Uses a donut chart to analyze total order amount by payment method.
- **Rating Analysis:** Uses a combination chart to compare order volume and average customer rating by city.
- **Cuisine Analysis:** Identifies the Top 5 Preferred Cuisines based on customer preferences.
- **Price Category Analysis:** Shows restaurant distribution across Budget, Mid Range, and Premium price categories.
- **Location-based Analysis:** Uses city-wise charts and map visualization to analyze restaurant and delivery performance geographically.
- **Delivery Performance:** Uses a Gauge Chart to display the Delivery Success Rate.
- **Order Status Analysis:** Uses a Funnel Chart to compare Delivered, Cancelled, and Rejected orders.
- **Interactive Slicers/Filters:** Provides City filters to dynamically analyze the dashboard based on selected locations.
- **Cross-Visual Interaction:** Selecting a city or category dynamically filters and updates the other visuals for comparative analysis.
- **Top-N Analysis:** Highlights the Top 5 Restaurants by Orders and Top 5 Preferred Cuisines.
- **Trend Analysis:** Uses a line chart to analyze restaurant growth based on opening year.
- **Conditional Formatting & Data Labels:** Highlights important values and makes comparisons easier across charts.





9. Insights & Conclusions

Key Findings:

- The dataset contains approximately 4K orders, 585 restaurants, and 820 customers.
- Slow delivery has the highest order volume with around 2K orders, followed by Normal delivery (~1.2K) and Fast delivery (~1K).
- Active restaurants are the largest restaurant-status category, with 350 active restaurants.
- Chennai has the highest number of restaurants (96), followed by Delhi (84) and Mumbai (82).
- The highest-order restaurants include Royal Biryani House, Dosa Corner, Urban Cafe, Spice Kitchen, and Pepper Dine.
- Male customers represent the largest customer group in the gender analysis.
- Chinese, South Indian, and Fast Food are among the most preferred cuisines.
- GPay and UPI contribute a significant share of the total order amount.
- The Delivery Success Rate is approximately 63%, indicating considerable scope for improving successful deliveries.
- Restaurant openings show fluctuations over the years, with a noticeable peak around 2018.

Descriptive Analysis

- Orders are distributed across Fast, Normal, and Slow delivery categories, with Slow delivery having the highest order count.
- Restaurant presence is highest in Chennai, Delhi, and Mumbai.
- The majority of restaurants are currently Active, while others are Closed or Temporarily Closed.

Diagnosis

- The high number of Slow deliveries suggests that delivery time may be an important operational issue.
- The relatively low delivery success rate indicates that cancelled and rejected orders are reducing overall delivery performance.
- The strong usage of digital payment methods suggests that customers prefer cashless payment options.

Predictive

- If Slow deliveries remain high, customer satisfaction and repeat ordering could potentially be affected.
- Increasing digital payment adoption can potentially lead to greater usage of UPI and mobile payment methods.
- Cities with lower restaurant penetration may represent opportunities for future restaurant expansion.

Prescriptive

- Improve delivery operations to reduce the number of Slow deliveries and increase the Delivery Success Rate.
- Consider expanding restaurant coverage in cities with strong customer demand but relatively fewer restaurants.
- Continue supporting UPI and digital payment methods, while monitoring payment preferences

10. Conclusions

The analysis provides an overall view of restaurant, customer, order, payment, and delivery performance. The key opportunity is to improve delivery efficiency and success rates while focusing on high-performing restaurants, popular cuisines, customer segments, and high-potential cities. The insights can support data-driven decisions related to restaurant expansion, customer engagement, marketing, and operational improvement.